Milestone 1: Theory and Data

CMP 3005 Term Project · MediDrone Istanbul · Team: [Member 1 · Member 2 · Member 3 · Member 4]

1. Problem definitions

Instance. A complete undirected graph $G = (V, E)$, $V = \{0, ..., n\}$ with depot 0; metric distances $d : E \to \mathbb{R}^+$; demands $q_i \in \mathbb{Z}^+$ with $q_i \le Q$; capacity Q; fleet size K. **Feasible solution**: at most K routes, each starting and ending at the depot, every customer on exactly one route exactly once, and $\sum_{i \in R} q_i \le Q$ for every route R.

Optimisation version: minimise $\sum_R \big(d(0, v_1) + \sum_{t=1}^{m-1} d(v_t, v_{t+1}) + d(v_m, 0)\big)$ over feasible solutions. **Decision version (CVRP-DEC):** given (G, d, q, Q, K) and B, is there a feasible solution of total distance $\le B$?

2. Reduction sketches

- **CVRP-DEC ∈ NP.** Certificate: the list of routes ($\le n + K$ numbers). The verifier checks that each customer appears exactly once, each load is $\le Q$, there are $\le K$ routes and the total cost is $\le B$, in $O(n + K) = O(n)$ time.
- **Hamiltonian Cycle $\le_p$ TSP-DEC.** Given $H = (V, E_H)$, build a complete graph with $w(u, v) = 1$ if $uv \in E_H$, else 2, and $B = |V|$. Building it takes $O(|V|^2)$. A Hamiltonian cycle gives a tour of cost $|V|$. Conversely, a tour of cost $\le |V|$ has $|V|$ edges of weight ≥ 1, so all have weight 1 and lie in E_H. Weights in $\{1, 2\}$ satisfy the triangle inequality, so the target is **metric** TSP. Our CVRP definition requires a metric, so this matters.
- **TSP-DEC $\le_p$ CVRP-DEC.** City 1 becomes the depot and the others become customers with $q_i = 1$, $Q = n$, $K = 1$, same d and B. A single route visiting all customers is exactly a Hamiltonian tour, and the costs are equal.
- **Composition.** The first reduction's output has polynomial size, so running the second on it is still polynomial. Hence HC $\le_p$ CVRP-DEC, so CVRP-DEC is NP-complete and the optimisation version is NP-hard.
- **Hidden problem: fleet feasibility.** Map Bin Packing (sizes s_i, capacity C, k bins) to customers $q_i = s_i$, $Q = C$, $K = k$, with arbitrary metric distances. The customers fit into K vehicles iff the items fit into k bins. So "can all customers be served with $\le K$ vehicles?" is (strongly) NP-complete. Even **finding a feasible solution** can be hard, which is why our Tabu Search allows overloaded intermediate solutions with an adaptive penalty.
- **Bosphorus metric.** With $d'(i, j) = e(i, j) + P[s(i) \ne s(j)]$: if i, k are on the same side, $d'(i, k) = e(i, k) \le e(i, j) + e(j, k) \le d'(i, j) + d'(j, k)$. If they are on different sides, j is on one of the two sides, so exactly one of the legs crosses and the right-hand side contains P. Then $d'(i, k) = e(i, k) + P \le e(i, j) + e(j, k) + P = d'(i, j) + d'(j, k)$.

3. Seed calculation and team instance

Seed = sum of the last four digits of all members' student IDs: **6936 + 2546 + 6698 + 4949 = 21129** → seed 21129.

The instance has $n = 100$ customers in 5 clusters plus 20 uniformly random customers. Capacity is $Q = 100$ and $K = \lceil \sum q_i / Q \rceil + 1 = 16$. Distances are Euclidean plus $P = 150$ when an arc crosses $x = 500$.

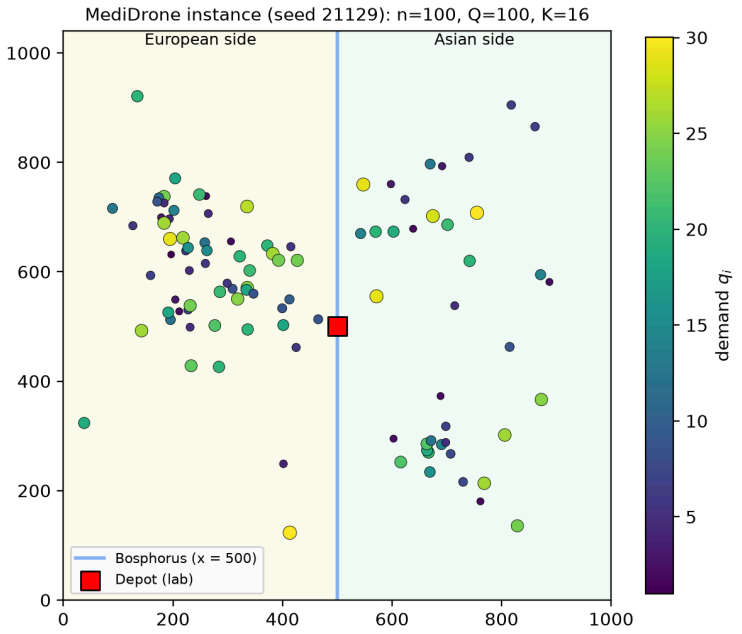


Figure 1: Our MediDrone instance. The depot is at the centre $(500, 500)$ and the Bosphorus line is at $x = 500$.

4. Tiny instances with exact optima

Instance	n	K	Optimum	Optimal routes
tiny-n7	7	4	3884.47	(4 3 1) (5 2) (7 6)
tiny-n8	8	3	3387.94	(6 7 5 2 1) (8 3 4)
tiny-n9	9	5	5835.41	(8 1) (2 3 4) (7 5 6) (9)

We compute the exact optima with a Held–Karp DP for every capacity-feasible subset ($O(2^n n^2)$), followed by a set-partition DP over subsets with at most K routes ($O(K3^n)$). For $n = 6$ the result matches a brute-force enumeration of all permutations and cuts (unit test).